Multi-Modal Non-Euclidean Brain Network Analysis With Community Detection and Convolutional Autoencoder

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***Abstract*—Brain network analysis is one of the most effective methods for brain disease diagnosis. Existing studies have shown that exploring information from multimodal data is a valuable way to improve the effectiveness of brain network analysis. In recent years, deep learning has received more and more attention due to its powerful feature learning capabilities, and it is natural to introduce this tool into multi-modal brain networks analysis. However, it would face two challenges. One is that brain network is in non-Euclidean domain, so the convolution kernel in deep learning cannot be directly used to brain networks. The other is that most of existing multi-modal brain network analysis methods cannot extract full use of complementary information from distinct modalities. In this paper, we propose a multi-modal non-Euclidean brain network analysis method based on community detection and convolutional autoencoder, which can solve the above two problems simultaneously in one framework (M2CDCA). First, we construct the functional and structural brain network, respectively. Second, we design a multi-modal interactive community detection method that exploits structural modality to guide functional modality for detecting community structure, then readjusts the nodes distribution so that the adjusted brain network can preserve the potential community information and is more suitable for convolution kernels. Finally, we design a dual-channel autoencoder model with self-attention mechanism to capture hierarchical and highly non-linear features, then comprehensively use two modalities information for classification. We evaluate our method on an epilepsy dataset, the experimental results show that our method outperform several state-of-the-art methods.**

***Index Terms*—Brain network analysis, community detection, convolutional autoencoder, structural modal, functional modal.**

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I. Introduction

THERE are billions of neurons and synapses in the human brain. Neurons connect and communicate with one another, and they work together to be responsible for human behavior, perception, and other tasks [1]. A number of relevant studies have shown that many brain diseases related to brain injury or cognitive dysfunction (such as Alzheimer's Disease, Epilepsy, and Autism) are root causes of damage to the internal nervous system of the brain [2], [3]. Therefore, the early detection of these injures and lesions is very important for the prevention and diagnosis of brain diseases in clinical practice. In recent years, neuroimaging technology has been developing rapidly, and various brain networks constructed forms provide us with great support for studying the internal organization and activities of the human brain. Therefore, brain network analysis methods that rely on neuroimaging technology have received more and more attention. In addition, many research works have indicated that among the methods for diagnosing brain diseases, the brain network analysis method is one of the most effective [4], [5]. Functional magnetic resonance imaging (fMRI) and diffusion tensor imaging (DTI) are two of the most commonly used brain imaging techniques. fMRI can reflect the time correlation between BLOD signals in brain regions, so the functional connection network (FC) of the brain can be constructed through fMRI data. DTI can reveal the physical connections between functionally related gray matter regions, thereby constructing the structural connection network (SC) of the brain. Many related works have proved that brain network analysis methods combining functional and structural data are more effective than using only single modality data for disease diagnosis [6].

The existing multi-modal brain network analysis methods can be divided into two categories. The first category is based on data fusion strategies, such as multi-view embedding, multi-kernel learning (MKL) and principal component analysis (PCA) [4]. However, these strategies only simply fuse the multi-modal data and cannot take into account the complementary information between the internal topological structures of brain networks constructed by two modalities. The second category is based on guiding strategy, using one modality to assist the other modality or constructing a unified brain network with multi-modal data [6], [7]. But in this way, for each subject, only one brain network can be obtained in the end, losing the unique

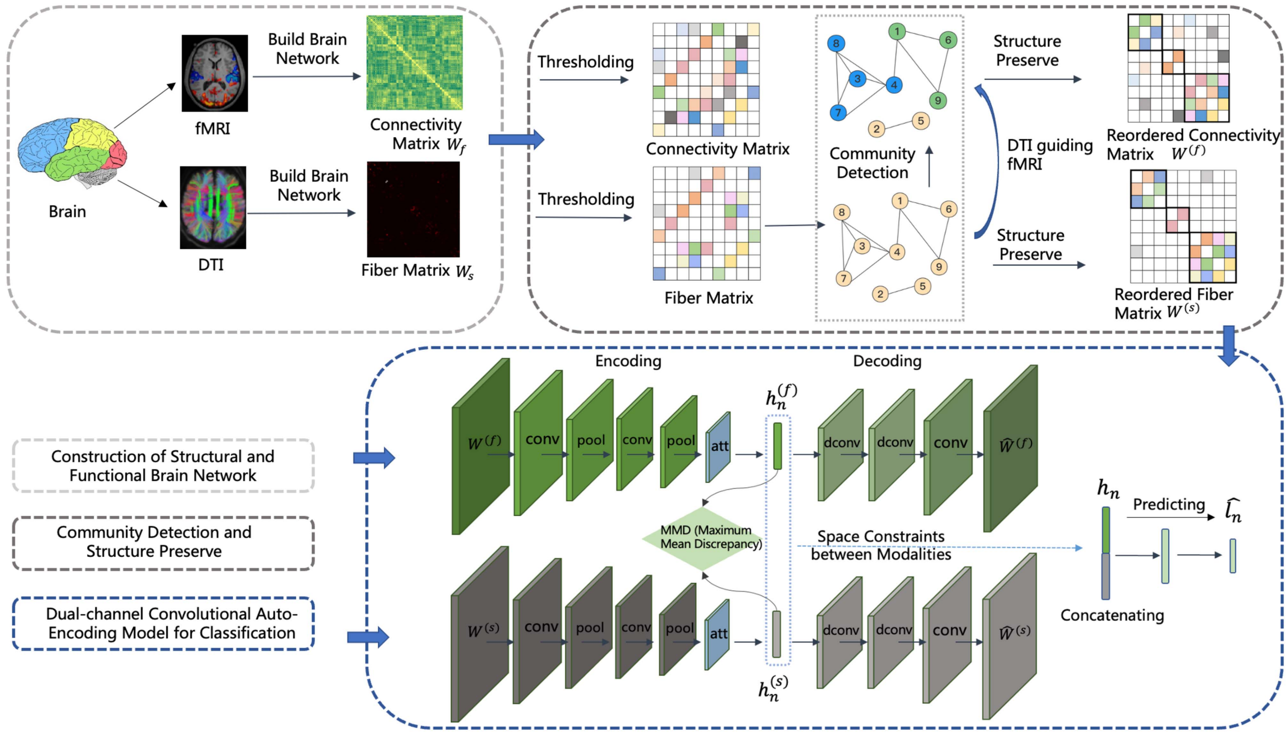

Fig. 1. The flowchart of our proposed method for multi-modal non-Euclidean brain network analysis with community detection and convolutional autoencoder.

characteristics of the brain network from each modality. In addition, these methods can only capture the shallow features from brain network, and ignore the hierarchical and highly non-linear relationships hidden in the typical graph structure data of the brain network. It has been demonstrated that some intrinsic structures inside the brain network, including small worlds or modules, and community structures, plays a vital role in the process of brain network analysis [8]–[10]. So it is important to preserve the structure information while brain network embedding. However, the aforementioned traditional multi-modal brain network analysis methods would fail to keep the essential structural information by converting the brain network matrix into a vector.

Many methods have been proposed to solve these difficulties. The subgraph mining and graph kernel method are both typical ones. Graph kernel methods usually calculate the similarity between two networks based on sub-structures, such as the shortest path, random walk [11], [12]. The graph kernel method can be used to integrate the local connectivity of the brain network, and the global topological properties. Nevertheless, the models based on graph kernel generally have shallow architectures, which means that they only can extract low order features and their feature learning ability in complicated situations is relatively weak, thus unable to effectively capture the potential, complex, and highly non-linear relationships in the brain network. The subgraph mining method aims to extract useful subgraph features from graph by using specific criteria [13], [14]. A widely used way is to select frequent sub-structure through connected subgraph patterns [15]. However, due to the huge complexity of brain networks, this method is difficult to accurately capture the highly non-linear characteristics. Moreover, it cannot detect the structural information inside the network. In recent years, deep learning methods are very popular, and have achieved good results in image processing and natural language processing [16], [17]. Many scholars try to introduce deep learning methods into medical image analysis, hoping to solve the limitations of traditional methods. For example, Jeon *et al.* used autoencoder technology to explore the rich representation of spatio-temporal patterns in rs-fMRI, and the most relevant features of the original features for diagnosis [18]. Hu *et al.* used restricted boltzmann machine (RBM) to mine potential sources in task fMRI data [19]. Li *et al.* used deep belief network (DBM) to learn important physiological characterization from fMRI data [20]. Deep learning methods are not suitable for directly using in brain network analysis because the connections between different brain regions are located in non-Euclidean domain [8], [9]. Therefore, most of existing brain disease diagnosis based on deep learning focus on neuroimaging rather than brain network. What's more, the research based on deep learning rarely pays attention to structural and functional modality simultaneously, while the majority of the studies only focus on either one.

In this paper, we propose a brain network analysis and disease diagnosis method based on community detection and dual-channel convolutional autoencoder technique. Fig. 1 gives a schematic illustration of our proposed method. We summarize the detailed algorithm proposed in our paper in Algorithm 1. First, we respectively construct the functional brain network as well as the structural brain network. Next, we design a multi-modal interactive community detection and structural preserve method, so that the potential module structure information in the network can be preserved and the adjusted brain network can be more suitable for convolution kernels. Then, the dual-channel convolutional autoencoder model with attention mechanism is designed to extract potential high-order information from two

Algorithm 1: The Detailed Algorithm Proposed of Our M2CDCA Method.

Input: fMRI data and DTI data; Parameters α, β, k;
Initialization: ϵ;
Construct $G_f = (V_f, E_f, \mathbf{W}_f)$ and $G_s = (V_s, E_s, \mathbf{W}_s)$;
Thresholding $\mathbf{W}_f$ and $\mathbf{W}_s$ as $\mathbf{W}'_f$ and $\mathbf{W}'_s$, respectively;
Perform multi-modal interactive community detection and structural preserve, and then obtain $\mathbf{W}^{(f)}$ and $\mathbf{W}^{(s)}$;
While not converged **do**:
  Calculate $\mathcal{L}_{sum}^{(t)}$ by using Eq. (6);
  Calculate the gradient of network parameters;
  Update network parameters using Adam optimizer;
  Calculate ACC;
  Check the convergence conditions $\mid \mathcal{L}_{sum}^{(t-1)} - \mathcal{L}_{sum}^{(t)} \mid \leq \epsilon$;
End While
Output: ACC.

modalities for classification. All in all, we conduct the data fusion of structural modal and functional modal from two aspects in our method. On the one hand, the strategy of structural modal guiding functional modal is adopted for brain network community detection and structural preservation rather than conducting community detection for two modalities respectively or only depending on one of two modalities. On the other hand, we design a dual-channel convolutional autoencoder model to capture high-order nonlinear features of structural and functional brain networks, and then concatenate the hidden feature vectors of two modalities for classification. The convolutional autoencoder model can well capture the highly non-linear feature of the brain networks. The dual-channel architecture facilitates comprehensively utilizing the complementary information of structural and functional modalities for classification. Through these two aspects of data fusion, the model can make full use of the complementary information between two modalities to improve the classification accuracies.

Our contributions can be summarized as the three folds:

1) We propose a dual-modal interactive community detection method, which fuses both the functional and structural data to learn a unified community division rule, avoiding the irrationality of obtaining two different rules for the same subject.
2) A dual-channel autoencoder model is designed to learn the high-order information hidden in two brain networks and improve classification accuracies. The convolutional autoencoder model can well capture the hierarchical and highly non-linear features of the brain networks. The dual-channel architecture facilitates comprehensively utilizing the complementary information of structural and functional modalities for classification.
3) A self-attention mechanism is introduced into the encoding processes for each branch to take account of the features that are more necessary for disease diagnosis.

The rest of this paper is organized as follows. In Section II, we summarize related works. Then, we present the proposed method in Section III. Section IV introduces materials used in the study and provides the experimental results on epilepsy dataset. In Section V, we give an analysis of the experimental results. Finally, we conclude our work in Section VI.

II. Related Work

A. Deep Learning Based Brain Network Analysis

The adoption of deep learning on brain network analysis has developed new kinds of methods to extract network features. Currently, the methods can be generally grouped into three types. Since GCN is more suitable for node classification rather than graph classification, most of the existing methods for brain network analysis based on GCN are to construct a graph using the brain networks of all samples as nodes, and then analyze the graph with GCN [21], [47]. For example, [21] and [22] are typical works. However, the definition of edges in a graph is a controversial issue in the GCN based methods. The edges in a graph reflect the relationship between two samples, which are generally correlated through age, gender or other information [21]. Some researchers question the credibility to judge whether the illness of a person is related to that of another, only depending on the basic information. The second type is to design a special convolutional method based on the characteristics of brain network, instead of directly using the traditional convolution kernel. For example, Kawahara *et al.* proposed novel edge-to-edge, edge-to-node, and node-to-graph convolutional filters to capture topological locality of brain network [45]. Mao *et al.* proposed a novel row-filter convolutional operator for brain network, which can effectively capture local pattern of graph by applying a row scanning on adjacent matrix [44]. As this method is based on a special convolutional strategy and the deep learning model must be designed for a specific brain network, it is not only cumbersome, but also has limited generalization performance. For example, a new convolutional operator is currently designed for structural networks. But when the target data is changed to functional networks, this method may not be applicable. In addition, because the convolutional method is newly designed, many traditional deep learning models cannot be used directly, the model must be specifically designed based on this convolutional method. The algorithms of the third type still use the traditional convolution kernel, but adjust the brain network structure. For example, Wang *et al.* introduced graph reordering and structural augmentation to enrich the spatial information for functional brain network [8]. Yan *et al.* incorporated the idea of node grouping into the design of neural network, so proposed a grouping-based interpretable neural network model [9].

B. Multi-Modal Brain Network Analysis

Among various neuroimaging data, fMRI and DTI are two most widely used modalities [7]. Early brain network research often uses single modal data to construct brain networks. In recent years, multi-modal network analysis that integrates structural and functional modalities has become more popular and achieved good results [6]. Multi-view embedding is an effective way to fuse information from multiple modalities [23]–[25]. For example, Ma *et al.* used a multi-view method to learn an unified network embedding from functional networks, structural networks, and central systems [23]. Zhang *et al.* fused multi-modal networks and longitudinal information to learn potential brain network representations [25]. But in multi-view method, each view is forced to map to a consistent representation, so that the different intrinsic properties of the network are ignored [10]. There are also some methods that comprehensively consider the information of multiple modalities in the construction of the brain network. For example, Yang *et al.* proposed a unified brain network construction method, which comprehensively utilizes the complementary information of functional modalities and structural modalities to construct a unified brain network [6]. But it can only consider the shallow features of the brain network, and cannot perform multi-modal fusion at a high level. Song *et al.* proposed a similarity-aware adaptive calibrated GCN (SAC-GCN), which can combine functional and structural information to predict SMC and MCI [21]. Yu *et al.* designed a novel framework based on multi-scale enhanced GCN (MSE-GCN), which fuses functional and structural information via the local weighted clustering coefficients [22]. However, both of the two methods use the brain networks of all samples to construct a hypergraph, and then classify the nodes of the hypergraph. Since the definition of edge in hypergraph is always controversial, the method is not as interpretable as our method in the field of brain disease diagnosis. Some works introduce deep learning methods into multi-modal brain network analysis tasks. For example, D′ Souza *et al.* proposed an integrated deep-generative framework, which jointly models complementary information from functional and structural connectivity to extract predictive biomarkers of a disease [26]. Huang *et al.* proposed an attention-diffusion-bilinear network for multi-modal brain network analysis to identify patients [4]. However, these methods do not adjust the structure of the brain network, and still cannot fundamentally solve the difficulty that the convolutional based neural network algorithm is not suitable for directly performing on the non-Euclidean brain network.

C. Self-Attention Mechanism in Brain Imaging Analysis

Attention is the ability by which the human brain processes visual information while also evaluating the relevance of input features. In the process of the development of attention mechanism, many variants have indeed appeared, such as local attention, global attention, hierarchical attention, self-attention and so on. However, since the self-attention mechanism allows input features to be the criteria for the attention itself, reduces the dependence on external information and is better at capturing the internal correlation of data or features, it has a wider application than other attention mechanisms and has shown better performance in many tasks. According to [51], the self-attention mechanism is suitable for the brain network analysis and brain imaging analysis due to its great ability to learn non-locally dependent information from data. For example, Wang *et al.* proposed a self-attention deep learning framework based on the transformer model on structural MR images from the ABIDE consortium to classify ASD patients from normal controls and simultaneously identify the structural biomarkers. The self-attention deep learning framework based on the transformer model can extract both local and global information from the input data, making it more suitable for the brain network data than the CNN-structural model [51]. Oh *et al.* combined self-attention mechanism with deep learning networks to extract both local and non-local information of higher importance from brain networks [52]. Yang *et al.* proposed a dual self-attention residual network that combines a spectrum attention module integrating local features with global features, with a channel attention module mining the interdependence between channel mappings to achieve better forecasting seizure performance [53]. Lei *et al.* proposed a dual aggregation network with self-attention mechanism to adaptively aggregate different information in infant brain MRI segmentation [54]. All these works prove that self-attention mechanism has a good application prospect in brain network analysis and brain image analysis.

III. Proposed Method

A. Functional Brain Network and Structural Brain Network Construction

Many complex relationships in real life can be abstracted into graphs, which can then be analyzed and studied using graph theory. The human brain, as one of the most complex systems in the world, can also be represented abstractly by graphs. Based on this theory, many graph-based methods for brain network analysis have also emerged.

In generally, a weighted graph can be represented as a triplet: $G = (V, E, \mathbf{W})$, in which V represent the node set of graph, E represent the edge set of graph, and $\mathbf{W}$ is a connectivity matrix, which can reflect the connectivity strength of each pair of nodes. Similarly, we abstract the brain network into such a triplet. The nodes in the graph correspond to the brain regions, and the edges in the graph correspond to the connections between different brain regions. In our experiments, we use two triplets $G_f = (V_f, E_f, \mathbf{W}_f)$ and $G_s = (V_s, E_s, \mathbf{W}_s)$ to represent the functional brain network and the structural brain network respectively. Among them, $\mathbf{W}_f$ is a functional connectivity matrix, which reflects the functional connectivity between different brain regions. It is obtained by calculating the Pearson correlation coefficient of time series data fMRI. $\mathbf{W}_s$ is a structural connection matrix constructed from DTI data, reflecting the number of nerve fibers between two brain regions.

B. Community Detection and Structure Preserve of Brain Network

1) Community Detection: Related works show that the internal community structure of brain network plays an important role in the diagnosis of brain diseases [9]. A community corresponds to a closely connected area in the brain, when the brain is damaged, the structure of communities would change obviously in the corresponding brain network [8]. In our work, we use spectral clustering to detect the community structure of the brain networks. The idea of spectral clustering is to divide the original graph into several small subgraphs so that the sum of edge weights in the subgraphs is as high as possible, and the sum of edge weights between subgraphs is as low as possible [27]. When applying spectral clustering on brain networks, the brain regions with higher correlation can be clustered into the same sub-graphs, while those with no correlation or lower correlation will be partitioned into different sub-graphs.

For brain network G, our goal is to divide the graph $G = (V, E, \mathbf{W})$ into k subgraphs: $A_1, A_2, \ldots, A_k$, which satisfies $A_i \cap A_j = \phi$, $A_1 \cup A_2 \cup \ldots \cup A_k = V$. Define the weight between any two subgraphs A and B as $W(A, B) = \sum_{i\epsilon A, j\epsilon B} w_{ij}$, the weight between k subgraphs as $W(A_1, A_2, \ldots, A_k) = \frac{1}{2}\sum_{i=1}^{k} W(A_i, \overline{A_i})$, in which $\overline{A_i}$ is the complementary set of A_i. In order to find appropriate sub-graph division to make the node weights in the sub-graphs are high, and the node weights between the sub-graphs are low, the optimization goal can be defined as

$$\min W(A_1, A_2, \ldots, A_k) = \frac{1}{2}\sum_{i=1}^{k} \frac{W(A_i, \overline{A_i})}{vol(A_i)}. \quad (1)$$

where $vol(A_i)$ represents the sum of edge weights in subgraph A_i. In order to simplify the calculation, according to [27], we can transform the above objective function to

$$arg \min_{\mathbf{F}} tr(\mathbf{F}^T\mathbf{D}^{-1/2}\mathbf{L}\mathbf{D}^{-1/2}\mathbf{F}) \quad s.t. \quad \mathbf{F}^T\mathbf{F} = \mathbf{I}. \quad (2)$$

where $\mathbf{L} = \mathbf{D} - \widetilde{\mathbf{W}}$, $\widetilde{\mathbf{W}}$ is the average value of connectivity matrix of all samples, $\mathbf{D}$ is degree matrix of $\widetilde{\mathbf{W}}$. After solving the characteristic matrix $\mathbf{F}$, we perform a traditional clustering on matrix $\mathbf{F}$, then the final brain network cluster division $C(c_1, c_2, \ldots, c_k)$ can be obtained.

2) Network Structure Adjustment Based on Multi-Modal Interaction: In the experiment, after using spectral clustering method to detect the community structure in the brain network, the rule for community division can be acquired. Then, according to this rule, the arrangement of brain regions in the brain network is adjusted to preserve the community structure and make it suitable for the convolution kernel. The specific adjustment method is: if the i-th brain region and the j-th brain region belong to the same community, then place them in the adjacent position of the brain network matrix. After that, the connectivity matrix of the brain network would be formed in an approximate block-diagonal fashion. Each diagonal block corresponds to one community, and the nodes belonging to this community are distributed in this diagonal block. The process of the adjustment is shown in Fig. 2. Fig. 2(c) is the primitive brain network, and through some community detection methods, the community structure shown in Fig. 2(d) can be found. Fig. 2(a) shows the original disordered brain network matrix, while Fig. 2(b) shows the brain network matrix that can reflect the internal community structure after node reordering.

For each subject, we collect two modalities and construct two brain networks. One is functional brain network $\mathbf{W}_f$ constructed by fMRI, and the other is structural brain network $\mathbf{W}_s$ constructed by DTI. Relevant studies show that community structures both exist in these two brain networks [10]. For two modalities, a direct approach is to detect the community structure and adjust the arrangement order of brain region of the two brain networks, respectively. But this approach will lead to two problems. One problem is that this approach can obtain two different community division rules for one subject, which is unreasonable. The other problem is that this approach does not take into account the interaction information between the two modalities. Inspired by [28], [29], we try to utilize the strategy of structural brain network to guide functional brain network.

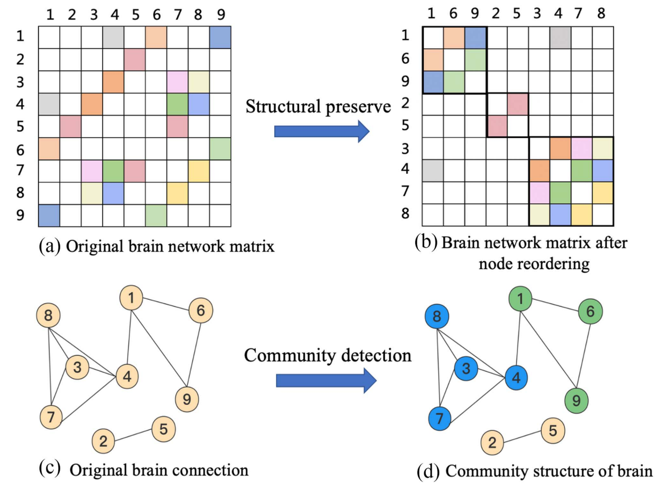

Fig. 2. A schematic of community detection and structure preserve of brain network.

Specifically, it can be divided into four steps: 1) First, we perform a threshold operation to transform the original structural brain network $\mathbf{W}_s$ and functional brain network $\mathbf{W}_f$ to $\mathbf{W}'_s$ and $\mathbf{W}'_f$, respectively. 2) Second, we utilize spectral clustering algorithm to detect the potential community structure in structured brain network, and then obtain the community partition rule. 3) Third, we adjust the order of network nodes in structural and functional brain networks respectively according to the community partition rule obtained from the previous step. 4) Finally, the adjusted functional brain network $\widetilde{\mathbf{W}'_f}$ and structural brain network $\widetilde{\mathbf{W}'_s}$ were fed into the dual-channel convolutional autoencoder model for learning the higher order important features and the complementary information of the two modalities to complete the brain diseases diagnosis task. It is worth noting that the community detection process and dual-channel convolutional autoencoder model training process are closely connected, and using different parameters in spectral clustering will get different community partition rule. During experiment, we use grid search strategy (includes the parameters of community detection process, as well as the parameters of the dual-channel convolutional autoencoder model) to find the most suitable community division rule, so that the rules take into account comprehensive information of both structural and functional brain networks rather than only depending on structural modal, and can significantly improve the performance of classication. For simplify, we substitute $\widetilde{\mathbf{W}'_s}$ as $\mathbf{W}^{(s)}$, substitute $\widetilde{\mathbf{W}'_f}$ as $\mathbf{W}^{(f)}$ to represent the structural and functional brain network respectively in the following introduction.

C. Dual-Channel Convolutional Autoencoder Model

1) Main Architecture: The autoencoder consists of two basic structures: encoder and decoder. Taking the functional brain network as an example, the functional brain network $\mathbf{W}^{(f)}$ is used as the input of the encoder, and the hidden layer feature $h^{(f)}$ is obtained after training. Then, this hidden layer is used as the input of the decoder, and the reconstructed functional brain network $\widehat{\mathbf{W}}^{(f)}$ is obtained after the decoder is trained. The encoding-decoding process of the structural brain network is the same as that of the functional brain network, and it is trained with functional brain network simultaneously. The original structural brain network $\mathbf{W}^{(s)}$ is sent to encoder to obtain the hidden feature $h^{(s)}$, and then the reconstructed structural brain network $\widehat{\mathbf{W}}^{(s)}$ is obtained by decoder. The more similar between original brain network and reconstructed brain network, the higher represented capabilities of hidden features. Therefore, the loss function in the encoding-decoding stage is defined as:

$$\mathcal{L}_{rec} = \frac{1}{N}\sum_{n=1}^{N}(\| \mathbf{W}^{(f)} - \widehat{\mathbf{W}}^{(f)} \|_2^2 + \| \mathbf{W}^{(s)} - \widehat{\mathbf{W}}^{(s)} \|_2^2). \tag{3}$$

The hidden layer of autoencoder model has a good representation learning ability, and it represents important high-order information of the input brain networks. In order to make comprehensive use of the information of the two modalities, we concatenate the hidden layer feature vectors of functional modal and structural modal, and then map the concatenated feature vectors to the low-dimensional space through a fully connected layer for classification. It is unreasonable to concatenate the two hidden feature vectors directly because they may be distributed in two different spaces. Therefore, we introduce Maximum Mean Discrepancy (MMD) constraint into the model to make the distance of the two spaces as small as possible. MMD can measure the similarity of two spaces, the larger the value calculated by MMD loss, the greater the difference in the distribution of the two spaces [30]. And if the calculated value is small enough, the two spaces can be considered in the same distribution [30], [31]. We hope that the hidden layers of the structural modality and the functional modality are nearly similar. Therefore, we define the MMD loss function as

$$\mathcal{L}_{mmd} = \frac{1}{N^2}\sum_{i=1}^{N}\sum_{j=1}^{N}(e^{-\|h_i^{(f)}-h_j^{(f)}\|_2^2} + e^{-\|h_i^{(s)}-h_j^{(s)}\|_2^2} - 2 * e^{-\|h_i^{(f)}-h_j^{(s)}\|_2^2}). \tag{4}$$

For classification, the cross-entropy loss is selected as the loss function due to its good property [32].

$$\mathcal{L}_{cls} = -\frac{1}{N}\sum_{n=1}^{N}[l_n log(\widehat{l}_n) + (1-l_n)(log(1-\widehat{l}_n))]. \tag{5}$$

Finally, the total loss function of the model is:

$$\mathcal{L}_{sum} = \mathcal{L}_{rec} + \alpha\mathcal{L}_{cls} + \beta\mathcal{L}_{mmd}. \tag{6}$$

where α and β are weight parameters. We aim to minimize the total loss function $\mathcal{L}_{sum}$ in the training process of the model.

2) Attention Mechanism: Attention mechanisms have been widely used in the recent deep learning studies [4]. They can attach each position in receptive filed with a different weight according to their importance, making it possible to focus more on important features and less on insignificant features, and that cannot be achieved by traditional convolution kernel. In particular, self-attention allows input features to be the criteria for the attention itself [48]. In our work, self-attention inspired by [33] is introduced to better capture hidden features. Fig. 3 shows the flowchart of self-attention mechanism used in our work.

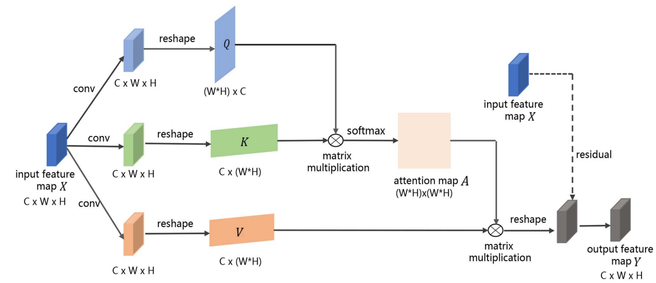

Fig. 3. The flowchart of self attention mechanism used in our work.

TABLE I
DETAIL INFORMATION OF THE EPILEPSY DATASET

Data	Quantity	Category	M/F [1]	Age Range	Mean Age
NC	114	right-handed	58/56	20-38	26.2
FLE	103	right-handed	53/50	17-51	24.1
TLE	89	right-handed	44/45	17-51	25.9

[1] M/F indicates Male/Female.

Specifically, let define the input feature map in the attention module as $\mathbf{X} \in \mathbb{R}^{C\times W\times H}$, where C, W, and H represents channel, width, and height dimension of feature map, respectively. Three convolutional layer with 1×1 kernel size are applied to feature map $\mathbf{X}$, and reshape the result matrix to obtain $\mathbf{Q} \in \mathbb{R}^{(W\times H)\times C}$, $\mathbf{K} \in \mathbb{R}^{C\times(W\times H)}$, and $\mathbf{V} \in \mathbb{R}^{C\times(W\times H)}$ matrix, respectively. Then, matrix $\mathbf{Q}$ and $\mathbf{K}$ are multipled and softmax is applied on the result matrix to generate the attention map $\mathbf{A} \in \mathbb{R}^{(W\times H)\times(W\times H)}$ [33]. The attention map is core of self-attention mechanism, whose entry $\mathbf{A}_{ij}$ represents the impact of the i-th position on the j-th position. Later, matirx multiplication is applied on $\mathbf{V}$ and attention map $\mathbf{A}$, then we reshape the result matrix to obtain output feature map $\mathbf{Y} \in \mathbb{R}^{C\times W\times H}$, which is with same dimension as the input feature map $\mathbf{X}$. Note that we learn from the idea of residual network rather than output $\mathbf{Y}$ directly [33], [34]. So $\mathbf{Y}$ is defined as

$$\mathbf{Y} = \gamma * \mathbf{Y} + \mathbf{X}. \tag{7}$$

where γ is a weight parameter and $\mathbf{X}$ is the initial feature map $\mathbf{X}$. At the beginning, γ is set to 0, the initial feature map $\mathbf{X}$ is output directly. Then γ is constantly updated in the process of model learning, so the output feature map $\mathbf{Y}$ which is weighted by attention feature map is obtained gradually. In our work, this attention layer is as the last layer of encoder to get hidden features with more powerful representation capacity.

IV. Experiments

A. Materials and Preprocessing

We collect raw rs-fMRI data and DTI data by Siemens Trio 3 T scanner from Jinling Hospital, Nanjing, China. There are a total of 306 subjects, including 114 normal controls (NC), 103 patients with frontal lobe epilepsy (FLE), and 89 patients with temporal lobe epilepsy (TLE). The detail information of the data set is shown in Table I.

We use the SPM8 in the DPARSF toolbox to pre-process all rs-fMRI images. Specifically, in order to obtain the initial functional time series, slice time are collected, corrected, rearranged, and normalized to the EPI template. Then, we utilize

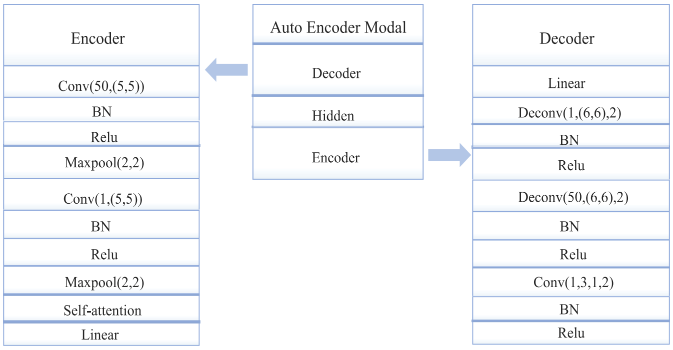

Fig. 4. The detail structure of dual-channel autoencoder model.

the de-trending to remove spurious sources of variance. Finally, we divide the resulting volumes consisting of 240 time points into 90 regions of brain interest (ROIs) by the AAL atlas, so that the information about brain activities can be reflected by those time series. We use the PANDA suite to process the DTI data. First, the FSL toolbox is utilized to correct the DTI distortion, remove the eddies, and extract the brain mask from the non-diffusion weighted (B0) image. Then, based on each subject's co-registered T1 images, the TrackVis is used to obtain fiber images by deterministic tracking method, and define anatomic areas using AAL conventions. Finally, the edge strength of the structural network can be naturally reflected by the quantity of fibers.

B. Experimental Setup

In order to evaluate the performance of our proposed method, we apply it to four different classification tasks, including NC vs FLE, NC vs TLE, FLE vs TLE, and NC vs (FLE and TLE). In our experiments, a dual-channel autoencoder model is designed to capture hierarchical and highly non-linear feature for classification. Furthermore, the model structure of each channel is same, which is shown in Fig. 4. Adam, an adaptive gradient-based optimization approach is adopted for parameter learning of model, in which weight decay is set to 0.05, and learning rate is dynamically adjusted according to the number of iterations. We perform a 4-fold cross validation and report the average results because the number of subjects in the datasets is limited. Classification accuracy (ACC) is used as an indicator to evaluate the classification performance.

C. Competing Methods

In order to verify the effectiveness of our method, several classical and state-of-the-art methods are selected to compare with our method. These methods fall into three categories: fMRI based methods, DTI based methods, fMRI and DTI based methods. More specically, fMRI based methods include PC [35], SGCN [36], WSGR [37], SSGSR [38], tHOFC [39], GAT [40], G-Unet [41], ConvRNN [42]. DTI based methods include GK [43], GCNN [44], BrainNetCNN [45]. fMRI and DTI based methods include MK [46], GCN [47], SAGPool [48], UBNfs [49], UM2BN [50].

D. Results on Epilepsy Dataset

Experimental results of all methods are summarized in Table II. As can be seen from Table II, in the four classification experiments, the accuracies of the proposed method are 78.45%, 80.56%, 74.19%, and 85.46%, respectively. Compared with other methods, the method in this paper achieves the highest accuracies in all four experiments: NC vs FLE, NC vs TLE, FLE vs TLE, and NC vs (FLE and TLE). More specifically, observing the Table II, we can draw the following conclusions.

First, the fMRI-based method achieves better results than DTI based method when using only one modality data. This indicates that in the task of diagnosing epilepsy diseases, fMRI data contains more effective information than DTI data. In general, the multi-modal method that combines fMRI and DTI data achieves higher classification accuracies than using only one single modality. This shows that the multi-modal classification method is indeed a promising method for the diagnosis of brain diseases.

Second, we chose some traditional machine learning methods (such as PC, WSGR, SSGSR, tHOFC, GK) and some deep learning methods (such as GAT, G-Unet, ConvRNN, SAGPool, GCN) as comparison methods. It can be seen from the experimental results that, in generally, deep learning methods achieves better results than traditional machine learning methods. The main reason for this is that, compared with traditional machine learning, deep learning methods can better capture potential high-order nonlinear features.

Third, whether compared with these traditional machine learning methods or deep learning methods, the method in this paper achieves the highest accuracies in all four tasks: NC vs FLE, NC vs TLE, FLE vs TLE, and NC vs (FLE and TLE). It can justify the rationality of our proposed community detection and structure preserve strategy, and the effectiveness of the dual-channel convolutional autoencoder model designed in our paper.

E. Analysis on Ablation Experiments

Relevant studies have shown that there are many potential community structures within the brain network, which can fully reflect the inside structural information and are of great significance for brain diseases diagnosis [9]. However, the distribution of brain region nodes in the original brain network is disordered, so this internal structural information cannot be well reflected. What's more, the convolutional based neural network methods are not suitable for brain network because brain network belongs to non-Euclidean domain. To address this problem, we design a multimodal interactive community detection and structure preserve method for brain network. Specifically, we adopt the strategy of structural modal guiding functional modal for community detection and obtain the community partition rule, then adjust the order of network nodes in structural and functional brain networks respectively to preserve potential structural information according to this rule. What's more, the adjusted brain networks are more suitable for convolution kernel. While designing the dual-channel convolutional autoencoder model, the Maximum Mean Discrepancy (MMD) is introduced to make the space distribution of hidden layer feature vectors of structural modal and functional modal as similar as possible.

TABLE II
PERFORMANCE OF PROPOSED METHOD AND COMPARATIVE METHODS

	Methods	NC vs FLE	NC vs TLE	FLE vs TLE	NC vs (FLE and TLE)
fMRI based methods	PC [35]	62.20	65.50	53.70	68.20
	SGCN [36]	67.32	74.50	70.51	69.34
	WSGR [37]	62.60	70.90	59.50	65.52
	SSGSR [38]	64.46	67.09	61.92	64.90
	tHOFC [39]	64.50	65.90	60.90	67.70
	GAT [40]	71.43	77.27	70.00	77.42
	G-Unet [41]	76.19	75.75	63.33	75.00
	ConvRNN [42]	74.86	72.69	72.25	76.09
DTI based methods	GK [43]	56.35	60.90	58.95	65.31
	GCNN [44]	54.86	60.88	57.00	64.60
	BrainNetCNN [45]	67.50	72.73	66.67	73.77
fMRI and DTI based methods	MK [46]	64.09	71.98	63.55	68.19
	GCN [47]	76.19	72.70	65.00	70.97
	SAGPool [48]	71.43	72.73	70.00	80.65
	UBNfs [49]	71.34	75.12	69.09	71.90
	UM2BN [50]	73.35	75.50	67.93	75.03
	Our Method	**78.45**	**80.56**	**74.19**	**85.46**

TABLE III
PERFORMANCE COMPARISON OF THE PROPOSED METHOD AND ITS VARIANTS

Methods	Community Detection	MMD Constraint	Self-attention	NC vs FLE	NC vs TLE	FLE vs TLE	NC vs (FLE and TLE)
M2CDCA-1				66.05	67.20	63.95	68.56
M2CDCA-2	√			71.50	73.90	69.95	76.20
M2CDCA-3		√		69.90	70.00	68.50	74.20
M2CDCA-4			√	70.30	71.50	68.40	74.90
M2CDCA-5	√	√		76.00	76.02	72.33	82.80
M2CDCA-6	√		√	73.00	74.20	70.04	77.36
M2CDCA-7		√	√	71.45	72.05	71.38	75.60
M2CDCA	√	√	√	**78.45**	**80.56**	**74.19**	**85.46**

Besides, we try to introduce the self-attention mechanism in this model to expect it can improve classification accuracies.

In order to verify the effectiveness of the three components of community detection, MMD constraint, and self-attention mechanism on our proposed M2CDCA method, we do some ablation experiments and present experiment results in Table III. In the table, M2CDCA-1,..., M2CDCA-7 represent the variants methods of our proposed M2CDCA method, and we use "√" to indicate that this component is used, otherwise it is not used. For example, M2CDCA-2 indicates that it uses the multimodal interaction community detection and structure preserve strategy, but in the classification, it only uses the dual-channel convolutional autoencoder model that does not introduce self-attention mechanism as well as MMD constraint. M2CDCA-3 indicates that it neither uses the community detection and structure preserve strategy nor introduces the self-attention mechanism, but directly sends the original brain networks to the dual-channel convolutional autoencoder model with MMD constraint for training. Compared with the accuracies of M2CDCA-5, M2CDCA-6, M2CDCA-7, and M2CDCA, we can find that abandoning any step of community detection, MMD constraint, and self-attention mechanism will reduce the experiment accuracies, and community detection is the most important of the three components, which has the greatest influence on the experiment accuracies. Compared with the accuracies of M2CDCA-2, M2CDCA-3, and M2CDCA-4 with the accuracies of M2CDCA-5, M2CDCA-6, and M2CDCA-7, it can be concluded that if any two components are discarded, the accuracies will be lower than one of them is discarded, which further illustrates the important role of those three components in improving the accuracies of disease diagnosis. In addition, from the comparison of the accuracies of M2CDCA-2, M2CDCA-3, and M2CDCA-4, it can still be shown that community detection is the component that has the greatest impact on the experimental accuracies among the three components.

The self-attention mechanism is better at capturing the internal correlation of features, so that the model with self-attention mechanism can focus on more important feature regions and obtain better classification accuracies. MMD can constraint the spatial distribution of the two hidden layer vectors of structural modal and functional modal as similar as possible, so that concatenating the two vectors for classification is more reasonable. However, the impact of these two components on the M2CDCA method is not as great as community detection. The main reason is that brain network belongs to the non-Euclidean domain, if there is no step of structure preserve based on community detection, the convolution kernel in the dual-channel convolutional autoencoder model cannot effectively perform feature learning on the brain networks. Therefore, although MMD constraint and self-attention mechanisms are added to the model, the impact of abandoning the community detection step on the experimental accuracies cannot be compensated.

F. *Comparison of Other Community Detection Methods*

In the experiment, we introduce the spectral clustering method to detect the potential community structure in the brain networks, and adjust the brain region order of the brain network matrix according to the results of community detection. In order to verify the rationality of the introduced spectral clustering method, other commonly used community detection methods, such as Kernighan Lin (KL) [55], Girvan Newman (GN) [56], Louvain [57] are all used to detect the community structure in brain networks, and compared with the spectral clustering

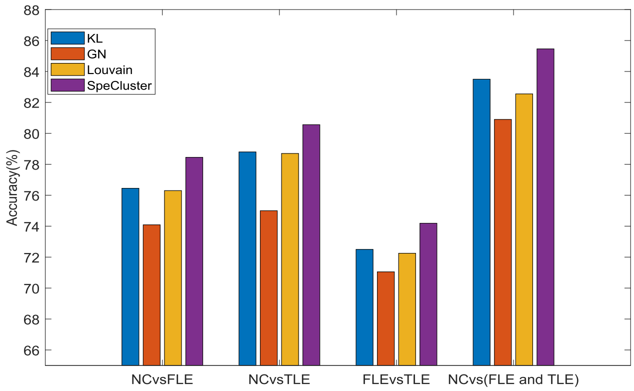

Fig. 5. Classification accuracies of four different community detection methods (Kernighan-Lin [55], Girvan Newman [56], Louvain [57], and Spectral Clustering [27] algorithm) on four different classification tasks (NC vs FLE, NC vs TLE, FLE vs TLE, NC vs (FLE and TLE)).

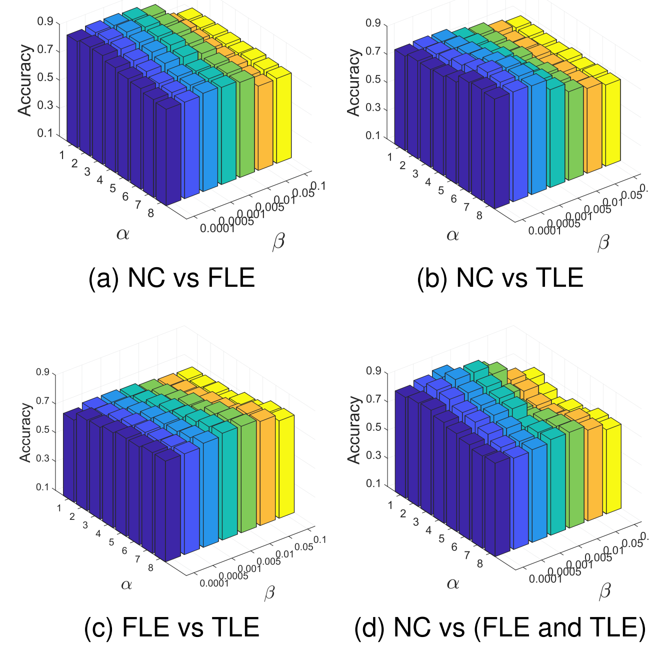

Fig. 6. The influence of parameter α and parameter β in the final loss function for all four classification tasks: NC vs FLE, NC vs TLE, FLE vs TLE, NC vs (FLE and TLE).

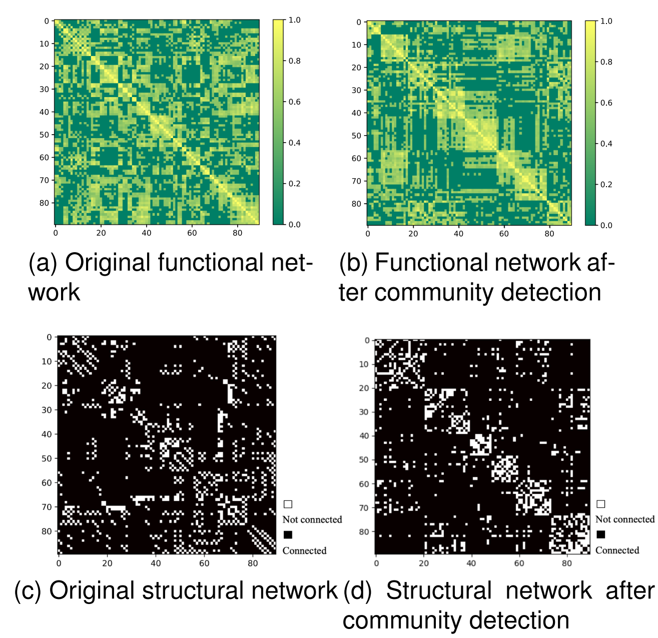

Fig. 7. The original brain network connection matrix, and the brain network connection matrix after community detection and structural information retention.

method. Specifically, we keep the same experimental settings and only change the community detection method. The experimental results are reflected in Fig. 5. It can be seen from Fig. 5 that compared with other community detection methods, the introduced spectral clustering algorithm achieves the highest accuracy in all four different experiments. This can verify the rationality and effectiveness of the spectral clustering algorithm in brain network community detection.

V. Discussion

A. Analysis of Parameter Sensitivity

As can be seen from Eq. (6), there are two parameters α and β in total loss function. We adopt the grid search method to study the influence of these two parameters on the experimental results, and find the optimal parameters. Specifically, we set α in range of {1, 2,..., 8}, β in range of {0.0001,..., 0.05, 0.1}, and reflect the experimental accuracies under each parameter combination in Fig. 6. It can be seen from Fig. 6 that, different parameter combination of α and β leads to different accuracies. On NC vs FLE task, the optimal parameter combination is: $\alpha = 2$, $\beta = 0.001$. On NC vs TLE task, the optimal parameter combination is: $\alpha = 6$, $\beta = 0.0005$. On FLE vs TLE task, the optimal parameter combination is: $\alpha = 8$, $\beta = 0.01$. On NC vs (FLE and TLE) task, the optimal parameter combination is: $\alpha = 2$, $\beta = 0.005$.

B. Structural Preserve of Brain Network

The brain network belongs to the non-Euclidean domain, the convolution kernels cannot be directly applied to non-Euclidean brain network. In addition, there are some community structures within the brain network, which are of great significance to brain disease diagnosis. In the experiment, we design a multi-modal interactive brain network community detection and structural preserve method. Specifically, we reorder the distribution of brain region nodes in brain network based on the results of community detection. We randomly select one sample and display the original brain network and the adjusted brain network matrix in Fig. (7). Among them, Fig. 7(a) is the original functional connection matrix, and Fig. 7(b) is the functional connection matrix after structure preserve. Fig. 7(c) is the original structural connection matrix. Fig. 7(d) is the structural connection matrix after the structure preserve. It can be seen that after adjustment, the brain network has an approximate block diagonal nature. One diagonal block corresponds to one community, and the brain region nodes belonging to this community are distributed with closer connections. After the adjustment, the brain region nodes

TABLE IV
PERFORMANCE COMPARSION OF OUR METHOD BASED ON DIFFERENT MODALITIES

	Methods	AUC	P	R	F1	ACC
NC:FLE	fMRI [58]	0.7045	72.34	74.55	73.45	74.00
	DTI [59]	0.6348	64.66	65.32	63.90	62.05
	fMRI+DTI	**0.7824**	**79.31**	**82.14**	**80.70**	**78.45**
NC:TLE	fMRI [58]	0.7580	68.50	70.05	69.45	74.67
	DTI [59]	0.7345	65.90	68.02	68.45	72.55
	fMRI+DTI	**0.8039**	**70.59**	**75.00**	**72.73**	**80.56**
FLE:TLE	fMRI [58]	0.6820	70.06	69.45	70.43	69.55
	DTI [59]	0.6645	68.09	67.05	68.90	67.45
	fMRI+DTI	**0.7324**	**76.00**	**73.08**	**74.51**	**74.19**
NC:(FLE & TLE)	fMRI [58]	0.7589	75.00	70.45	74.06	73.99
	DTI [59]	0.7368	72.00	68.89	68.20	68.90
	fMRI+DTI	**0.8819**	**93.94**	**83.78**	**88.57**	**85.46**

with stronger correlation belong to adjacent positions in the brain network matrix, so it is more suitable for convolution kernels.

Note that for functional brain networks, the correlation between brain regions is calculated by Pearson correlation coefficient and normalized, so it is evenly distributed between [0,1] and Fig. 7(a) (b) are shown in color. For structural brain networks, the correlation between brain regions indicates the number of physical fibers between two brain regions. Since the number of physical fibers between some brain regions is 0, and the number of physical fibers between some brain regions is very large, we normalized the original structural brain network into the interval [0, 1], and selected 0.5 as the threshold. If the value is greater than 0.5, it is considered to be connected and set to 1. If the value is less than 0.5, the connection is disconnected and set to 0. Therefore, for structure brain networks, binary heat maps can represent their internal connections.

C. The Effectiveness of Multi-Modal Fusion

In the experiment, we comprehensively use the information of structural modal and functional modal to achieve better performance. In order to verify the effectiveness of the fusion of two modalities, we do some comparative experiments, and use AUC, P, R, F1, ACC as evaluation indicators. Suppose that TP represents the number of patients predicted correctly, FP represents the number of healthy people predicted to be patients, TN represents the number of healthy people predicted correctly, and FN represents the number of patients predicted to be healthy. Indicators P, R, F1 can be defined by

$$P = \frac{TP}{TP + FP}. \tag{8}$$

$$R = \frac{TP}{TP + FN}. \tag{9}$$

$$F1 = \frac{2 \times P \times R}{P + R}. \tag{10}$$

AUC represents the sum of the areas of each part under the Receiver Operating Characteristic (ROC) curve. The experimental results are shown in Table IV. It can be seen from the Table IV that, no matter which classification task or which evaluation indicator, the experimental results obtained using only single modality data are inferior to using multi-modal data comprehensively. This can prove the effectiveness of multi-modal data fusion strategy [60], [61]. In our method, we conduct the data fusion of structural modal and functional modal from two aspects. On one hand, we utilize a multi-modal interactive community detection method to retain the internal structure information of the brain network. On the other hand, we design a dual-channel convolutional autoencoder model with MMD constraint, which can well integrate the high-order information of two modalities for classification. Through the above two aspects, the complementary information between modalities can fully mined to improve the classification accuracies.

VI. CONCLUSION

In this paper, we propose a brain network analysis and disease diagnosis method based on community detection and dual-channel autoencoder model. It overcomes the problem that the convolution based neural networks cannot be directly applied to brain networks in non-Euclidean domain, simultaneously effectively integrates complementary information from different brain network modalities to promote diagnosis. First, we construct functional brain network as well as structural brain network. Then, we design a multi-modal interactive community detection method that uses structural modal to guide functional modal for detecting the community structure in the network and readjust the distribution of network nodes to retain the potential structural information of the brain networks. Finally, a dual-channel autoencoder model with self-attention mechanism is designed to capture hierarchical and highly non-linear feature, then comprehensively use the information of the two modalities for classification. The experimental results show that our method can achieve promising performance on the epilepsy dataset.

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